Identification of an Unknown – "A"

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Introduction

The goal of this experiment was to identify an unknown compound by studying graphs produced by different compound-identifying technologies. There are many practical applications and useful reasons to identify unknown compounds, including to ensure safety, determine the quality of products, and identify any possible pollutants or contaminants.

Body

In this experiment, three different analyzing technologies were used to identify the unknown compound. The first, gas chromatography-mass spectrometer (GC-MS) is a combination of a gas chromatograph and mass spectrometer, which is used to find the retention time of each compound in a mixture. This technology is useful to identify solid, liquid or gaseous samples. With this method, a sample is vaporized and separated into its various components using a capillary column coated with a stationary phase. As the components are separating, each compound eludes from the column at different times according to its boiling point and polarity (retention time). This can then be used to find the identity of each compound in the mix. After leaving the column, the components are ionized and fragmented in the mass spectrometer. This process is done using wither electron or chemical ionization sources. The ionized molecules are then moved through a mass analyzer and separated based on their ratio of mass to charges (m/z). The final steps are when the ions are analyzed using their m/z ratios. On the graph, peak areas generate a unique mass spectrum that can be used for compound identification.

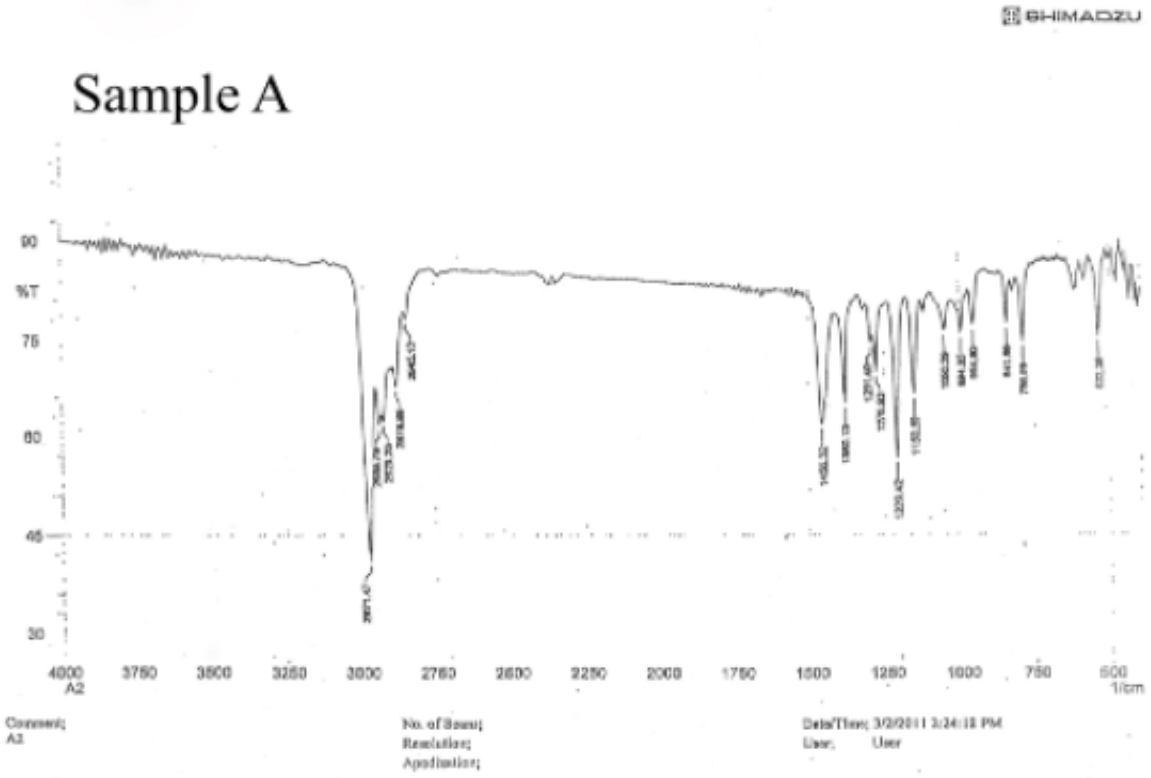
The second technology, Nuclear Magnetic Resonance Spectroscopy (NMR) involves the study of the interaction between electromagnetic radiation and the nuclei of atoms. In this process, a sample is placed in a glass vial and put inside an electronic coil, which itself sits inside a powerful magnet. The magnet causes the nuclei of the atoms within the ample to align to its

field, which cause them to rest in a consistent alignment. NMR is a especially useful for compounds with an odd number of protons or neutrons because their nuclei have built-in magnetic moments and angular momentum that give them “spin.” As long as this magnet is strong enough, then the nuclei will align their spins with the magnetic field. The electronic coil releases a series of radio-frequency pulses at a certain frequency in order to excite each specific nuclei, then detects how much energy is released as the nuclei return to their resting positions. This technology is used in order to determine the structure of a compound.

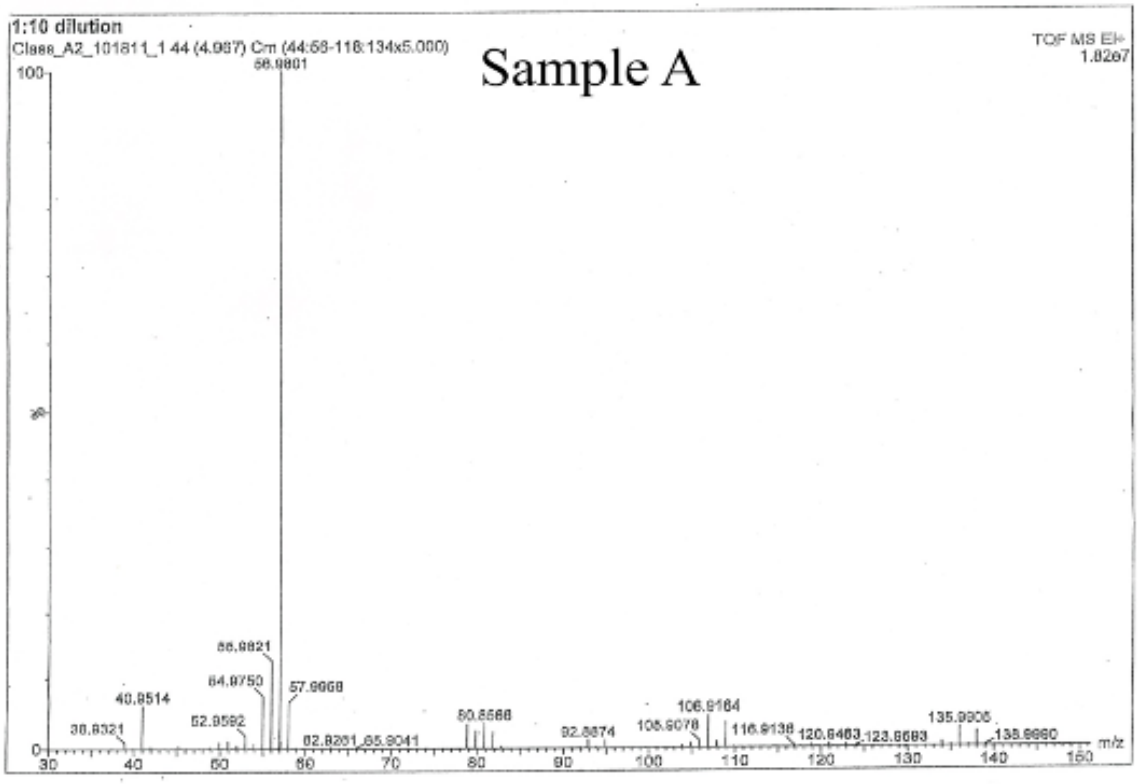
Finally, the Fourier Transform Infrared Technology (FTIR) is a device used to find the infrared spectrum of absorption of different molecules. This instrument utilizes an interferometer, which itself consists of a source, a beam splitter, two mirrors, a laser and a detector. This helps to provide information about the functional groups present on that compound. The energy moves from the source to the beam splitter, which divides the beam in half. One-half of this beam is transmitted to a moving mirror, while the other gets transmitted to a stationary one. The calibrating laser controls the constant velocity at which the moving mirror is disturbed. The two beams are reflected off the mirrors and come together at the beam splitter, which creates an interference pattern. This pattern is then transmitted through the sample compartment to the detector, and the signal is subjected to the Fournier Transform function and a spectrum is generated.

Data

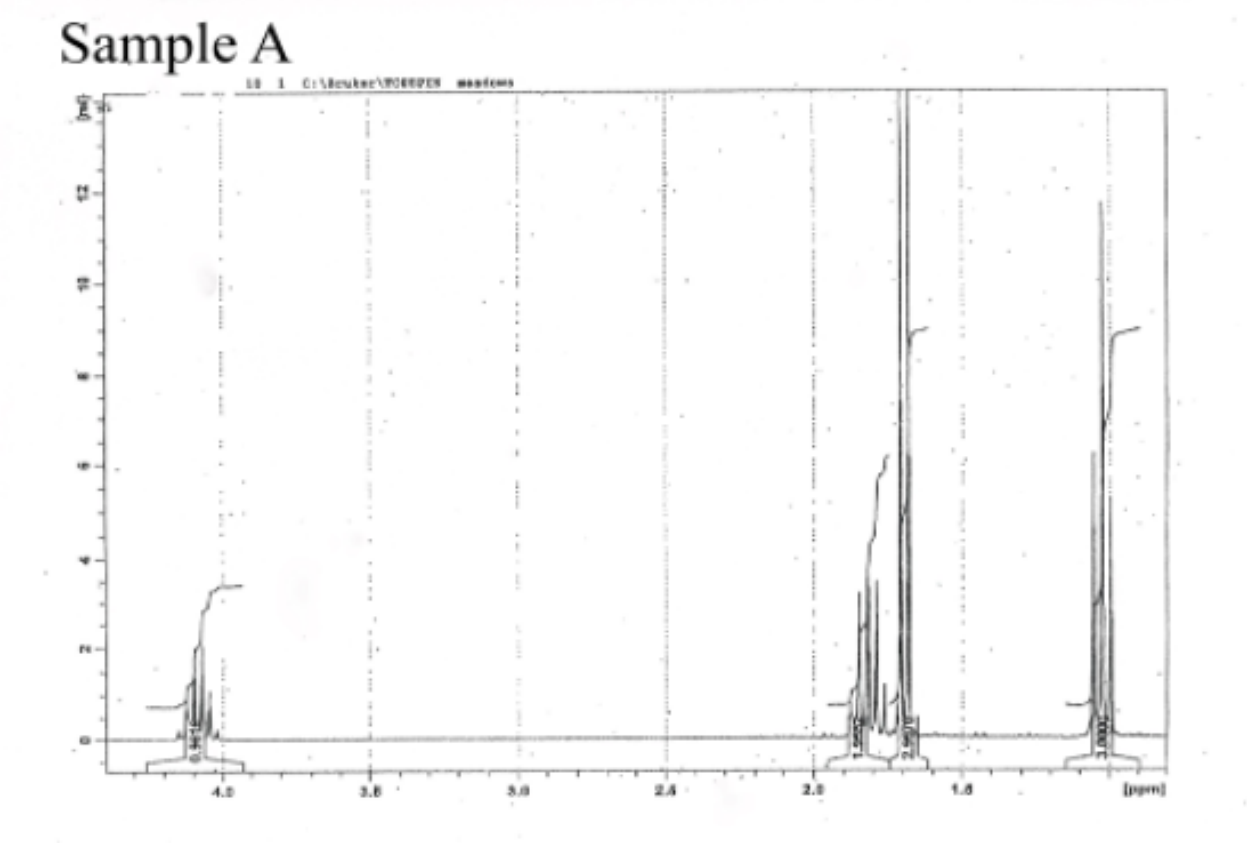
FTIR Sample A



GCMS Sample A



NMR Sample A



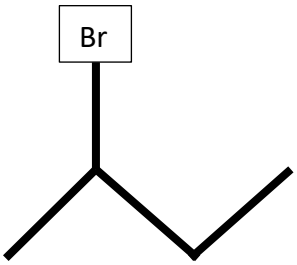
Analysis

Based on the graphs of spectra attached above, the unknown compound is most likely 2-bromobutane. Based on the FTIR graph, it can be observed that the lines between 2971-2845 cm^{-1} indicate C-H stretching of an alkane, and the marks at both 1458 cm^{-1} and 785 cm^{-1} indicate C-H vibrations and C-Br stretching, respectively. When looking at the GCMS graph, m+ is 136 and m+2 is 198, indicating the presence of a halogen. Finally, in the NMR spectrum, the 4.1-4.3 lines show a sextet, indicating a carbon with hydrogen bonded to it. Going toward the right, the next area of interest between 1.8-1.9 show a pentet, signifying a carbon with two hydrogens bonded to it. The next jump in the graph at 1.7 shows a doublet, indicating a carbon with three hydrogens bonded to it, and the last mark at 1.0 shows a triplet, indicating a carbon with three

hydrogens bonded to it. All of these graphs combined leads to the conclusion that the molecular formula for this compound should be C_4H_9Br, and they also indicate that in the structure the bromine should be at locant #2. This leads to the identity of the compound being 2-bromobutane.

Conclusion

Based on the analysis of all three graphs above, it can be inferred that the identity of the unknown compound is 2-bromobutane. The chemical structure for this compound is most likely as drawn below.



References

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